Solar System Models · Pupil Resources

Learning goal: I can use solar system data and a scale model to link distance from the Sun to orbit time.

Arrival Questions (In your book)

1. Which planet is closest to the Sun?

2. How many days does Earth take to orbit the Sun once?

3. Which takes longer to orbit the Sun: Mercury or Jupiter?

4. Can you ask an adult to help?

Starter Question

Why does Neptune take 165 Earth years to orbit the Sun while Mercury takes only 88 days?

A. Neptune is much further from the Sun, so it has a much longer orbit path and weaker solar pull.

B. Neptune stops moving in winter.

C. Mercury is bigger than Neptune.

Method Strip

1. Lay the 100 cm paper strip flat on your desk and label 0 cm as THE SUN.

2. Mark the inner rocky planets at 1 cm (Mercury), 2 cm (Venus), 3 cm (Earth), and 5 cm (Mars). STOP AND CHECK.

3. Mark the outer giants at 17 cm (Jupiter), 32 cm (Saturn), 64 cm (Uranus), and 100 cm (Neptune).

4. Compare your strip spacing with the Orbit Time column in the data table.

5. Write the distance-to-orbit-time pattern and one model limitation in your book.

Exit Ticket

Which planet takes longer to orbit the Sun: Earth (365 days) or Neptune (165 years)?

State the pattern linking a planet's distance from the Sun to its orbit time.

Give one strength and one limitation of our 100 cm paper strip model.